

Efe-Oghene Erhie

Mechanical engineer moving into robotics software. MEng candidate at Waterloo.

[LinkedIn](#) [GitHub](#)

Education

MEng, Mechanical & Mechatronics Engineering — University of Waterloo

Co-operative, course-based · September 2026 – present

BSc, Mechanical Engineering — University of Manitoba

Graduated June 2022 · GPA 4.36 / 4.50 · Best Graduating Student, Department of Mechanical Engineering

Experience

Manufacturing Specialist — Antec Controls

November 2022 – August 2026

- Operated and maintained automated test equipment for airflow testing of valves used in critical environments.
- Owned technical documentation and calibration procedures maintaining NVLAP accreditation for airflow measurement systems.
- Supported planning and execution of a production facility relocation into a space twice the size of the previous one.
- Provided strategic support implementing a new ERP system (IFS Cloud) integrating manufacturing and inventory processes.

Operations Leadership Trainee — PepsiCo Beverages Canada

August 2022 – October 2022

- Led a project to improve reliability of critical equipment at the beverage production plant.
- Coordinated equipment contractors to maximise the value of on-site visits.

- Analysed and improved the changeover process on key machines of the bottling line.

Content / Study Skills Tutor — Academic Learning Centre, University of Manitoba

August 2021 – May 2022

- Tutored engineering and science courses including thermodynamics, fluid mechanics, and CAD.
- Coached students on study technique and time management to build independent learning capability.

Production Shift Supervisor (Co-op) — PepsiCo Beverages Canada

January 2021 – September 2021

- Directed a production team against cost, line efficiency, waste, safety and productivity standards.
- Oversaw facility equipment reliability to minimise downtime.
- Built front-line capability through training and communication.

Technical Skills

- **Languages** — Python, C++, Rust, JavaScript
- **Simulation & numerical** — NumPy vectorisation, spatial partitioning, control-loop tuning, cross-implementation performance profiling
- **Systems** — Rust/WebAssembly toolchain, Tauri, event-sourced data models, static-site architecture
- **Engineering** — SolidWorks, AutoCAD, Fusion 360, GD&T, automated test equipment, calibration procedure authorship, NVLAP-compliant documentation, ERP systems (IFS Cloud)

Selected Projects

Full write-ups and runnable demos are on this site.

Boids flocking simulation

Python · NumPy · Rust · WebAssembly

A flocking model built three times over to study the same system from different angles: a scalar Python reference, a vectorised NumPy rewrite, and a Rust core compiled to WebAssembly. Covers control-loop stability under changing gains, and the move from an $O(n^2)$ neighbour scan to a spatial grid.

omni-me

Rust · Tauri · Dioxus

A cross-platform personal-data application. Event-sourced transaction model with multi-source reconciliation, recurring-transaction detection, and multi-currency account handling.

mylearnbase

Zola · Tera · JavaScript

This site. Build-time TF-IDF for content-based related posts, in-browser search over a pre-built index, and self-contained interactive demos embedded in articles.

Awards

- Best Graduating Student, Department of Mechanical Engineering — 2022
- Dean's Honour List, Faculty of Engineering — 2018 to graduation
- UMSU Scholarship — 2019 to 2021
- International Undergraduate Student Scholarship — 2019 to 2021
- Grettir Eggertson Memorial Scholarship — 2019 to 2021